

The 1997-98 El Niño: A monthly and seasonal snapshot of primary productivity off the coast of Louisiana, United States of America.**1. Introduction:**

In 1997–98, one of the strongest El Niño events recorded to date occurred, and its effects on weather and climatic regimes were experienced on both intermediate and large scales (Rosenzweig and Hillel, 2008). An area of interest was the Gulf of Mexico, particularly the region off the southeast coast adjacent to the Mississippi Delta in Louisiana, United States. In this report, we briefly examine the bathymetric properties of the shelf and the monthly and seasonal climatological snapshots of the area during the 1997–98 El Niño event.

According to Rabalais and Turner (2019), the shelf off Louisiana's Mississippi Delta is considered one of the largest artificially induced hypoxic bottom-water zones due to substantial discharges of nutrients such as nitrogen (N) and phosphorus (P) from the upper and middle reaches of river catchments, including the Mississippi River, into the delta and subsequently the shelf. This system was reported to contribute approximately 80% of the annual freshwater flow into the ocean, as well as 91% of the estimated annual N load and 88% of the P load between 1972 and 1993 (Dunn, 1996; Rabalais and Turner, 2019). This shelf is important for several reasons, including the socio-economic benefits derived from the local commercial fishing industry, which depends on this productive and lucrative crawfish fishery that dates to the 1800s (McClain and Romaine, 2004; Gambill and Midway, 2024).

For important inshore fishery resources such as crawfish, which depend on optimal dissolved oxygen concentrations in the water, harmful algal blooms occur when microalgal populations photosynthesise rapidly under conditions of high nutrient availability. This increases the carbon load available to detritivore organisms, which respire and consume large amounts of dissolved oxygen, thereby suffocating gill-breathing fauna and causing high mortalities (Overstreet and Hawkins, 2017).

Seasonal variations in dissolved oxygen concentrations in bottom waters along the Louisiana shelf are largely driven by anthropogenic nutrient inputs in the region (Rabalais and Turner, 2019). These waters are predominantly influenced by mesoscale eddies formed by inflowing water entering through the Yucatán Channel into the Gulf, where they are subsequently perturbed by the opposing Gulf-Loop Current before converging and exiting through the Florida Strait into the Gulf stream (Hurlburt and Thompson, 1982; Candela et al., 2019). Favourable prevailing southeasterly winds blowing from April to August (peak spring-winter season) encourage physical interactions between the Louisiana-Texas shelf slope and mesoscale eddies, thereby initiating upwelling events off the coast.

The Gulf-of-Mexico is characterised by high chlorophyll-a concentrations along the inshore coastal margin throughout the year (Zavala-Hidalgo et al., 2006). It has been reported that chlorophyll-a concentrations are lower during summer compared to autumn and winter, when concentrations increase due to localised climatic events such as cold fronts (Rabalais and Turner, 2019). Interest in this area stems from investigating how the 1997–98 El Niño event influenced variations in chlorophyll-a concentrations along the coastal margin of Louisiana, United States. The case study will be centred around the Mississippi Delta area (Figure 1).

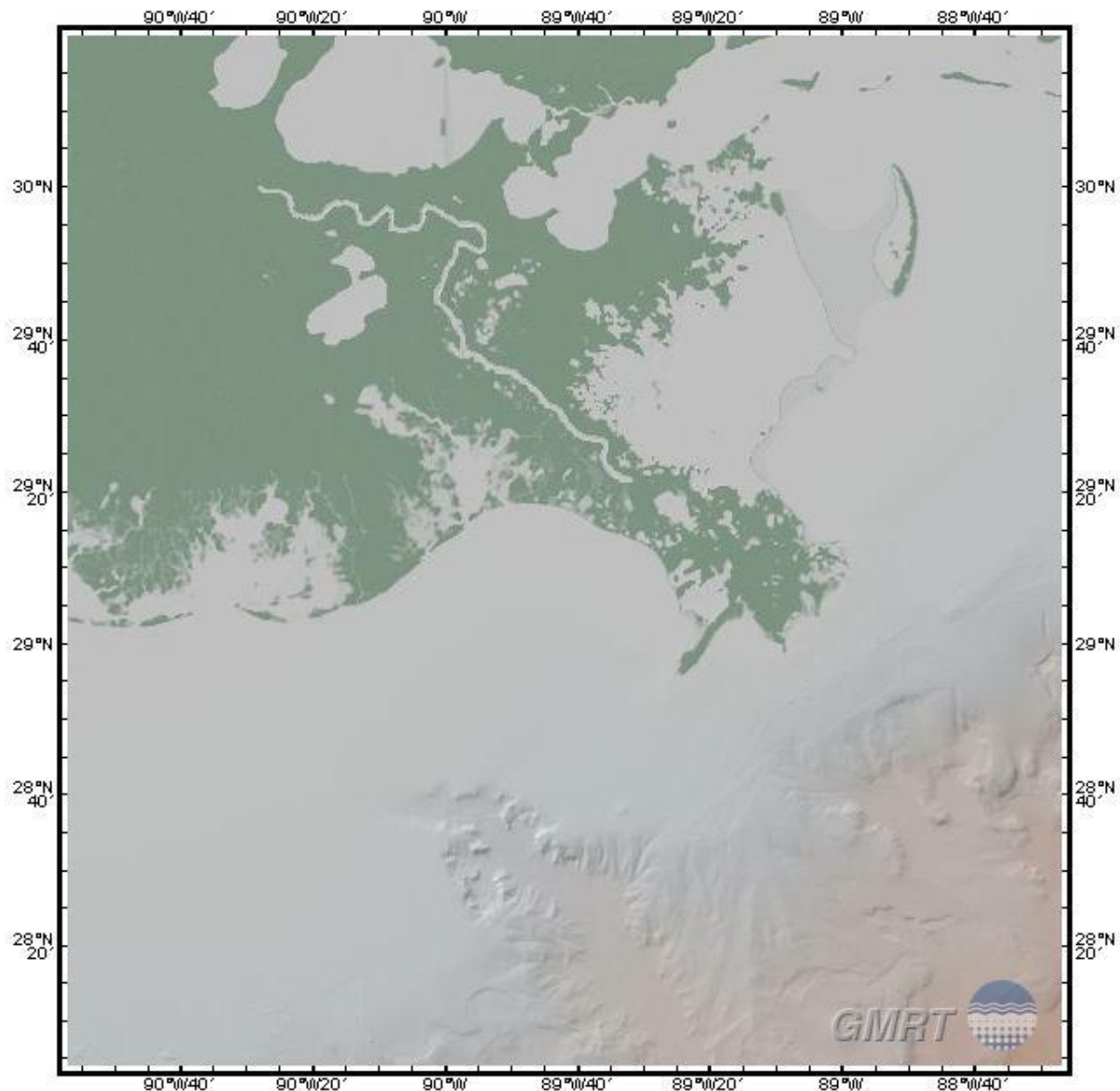


Figure 1: Aerial representation of the Mississippi Delta area off Louisiana, US.

2. Description of quantitative data and metadata:

2.1. Bathymetry:

Global Multi-Resolution Topography data synthesized by Ryan et al. (2009), was used for this analysis. The data was extracted with a cylindrical equidistant projection. The data

had altitude values with values under water being recorded as negative. Therefore, only negative datapoints were used for the analyses. The format of data was downloaded as a Coords/ CF compliant NetCDF data file with a grid resolution of 344.60 m/node with each node being 1139 nodes wide and 1180 nodes high. The bounding box was 250 by 250 km in area and then coordinates were 30.33° and 28.07° N latitude and then 88.45° and 90.95° W longitude.

The shelf off the coast of Louisiana is very shallow, ranging between 0 and 50 m in depth, and extends far offshore. However, the shelf narrows towards the peninsula. At the shelf break, the gradient becomes steep, with some underwater canyons visible from approximately 60 m below the surface (Figure 2). This shelf is important for ecosystems such as reefs, which host entire marine communities and underpin the trophic structure and interactions among ecosystem components that support local fisheries, ecological balance, and ecosystem health. With such shallow depths, a larger proportion of the surface water column, relative to offshore waters, is exposed to visible light and therefore solar energy. This enhances conditions for phytoplankton production, including chlorophyll-a, particularly in areas where productivity is supplemented by nutrient inputs from terrestrial sources such as the Louisiana Delta. This means that inshore marine ecosystems along this coastline are susceptible to eutrophication due to the combined influence of these factors.

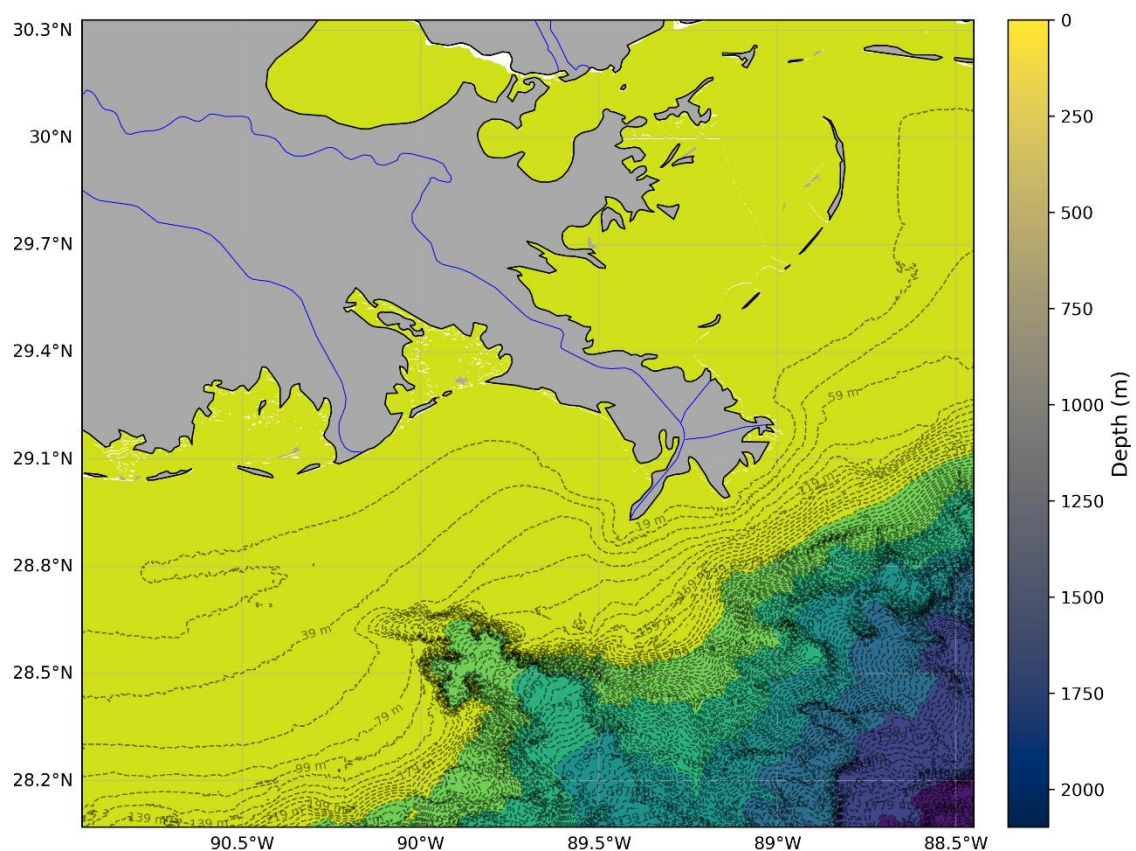


Figure 2: Bathymetry of the ocean floor off the Louisiana coastline, United States of America.

2.2. Chlorophyll-a:

The chlorophyll-a data used for the purpose of this report was extracted for the 250 × 250 km area encompassing the Mississippi delta coastline. The satellite measured chlorophyll-a concentrations over the surface of the ocean and data points were grouped in a monthly fashion (September 1997 – August 1998). This allows for climatological analyses over seasons to determine variations in chlorophyll-a concentration as a measure of productivity within an area. The dataset was extracted from the ESA Climate Change Initiatives (CCI) Ocean Colour Product (Version 5.0), produced by the Plymouth Marine Laboratory (PML). The product includes global coverage of level 3 monthly climatology of chlorophyll-a concentration variable with a resolution of 4 km. The satellite sensors used to collect data were SeaWiFS, MODIS-Aqua, MERIS, VIIRS, and OLCI.

3. Region Context

This study area is located over the northern continental shelf of the Gulf of Mexico and captures strong environmental variability driven by the Mississippi River Delta. This includes river plume expansion and retraction, strong seasonal chlorophyll gradients, shelf-current instability, and episodic hypoxia (“dead zone”) development (Jutras, 2023).

4. Data Analysis

The annual average chlorophyll-a concentrations are highest inshore, where they range between 0.60 and 1.23 mg.m⁻³ (Figure 3). These concentrations decrease rapidly offshore and are concentrated predominantly over the continental shelf. However, the maximum average chlorophyll-a concentration remains below 2 mg m⁻³, which falls within the typical range of productive coastal waters (1–10 mg.m⁻³) (Morel et al., 2010), while the mesotrophic waters indicate relatively barren areas of chlorophyll-a primary production (Figure 4). Overall, little seasonal variation in chlorophyll-a concentrations was observed on the Louisiana shelf.

Similarly, the monthly average chlorophyll-a concentrations show relatively little variation throughout this particular year of data (Figure 4). However, a few points on the plot display higher-than-normal levels associated with eutrophication events. These may have been triggered by the concurrent El Niño event, which causes increased rainfall over land, resulting in greater freshwater discharge into the ocean. This freshwater carries additional nutrients, which then become available for phytoplankton photosynthesis and carbon production (Halsey and Jones, 2015). However, no bloom events occurred, as evident from the monthly climatology. This is likely due to a balance between the warm current system present in the area (Leipper, 1970) and the relatively shallow continental shelf, which is not deep enough to generate sufficient momentum for prolonged upwelling events. Consequently, the system depends heavily on nutrient input from the Mississippi Delta.

At the point of interest, where relatively high chlorophyll-a concentrations were recorded (Figure 5), the overall mean concentration was higher than the regional average across the study period. Interestingly, between September and October 1997, both the regional mean and the point-of-interest mean decreased concurrently. However, while the regional mean increased gradually until May 1998, concentrations at the point of interest increased rapidly from approximately 1 mg m^{-3} to 1.4 mg m^{-3} by February 1998. Furthermore, as the regional mean decreased towards the end of August 1998, an increase in chlorophyll-a concentrations was still observed at the point of interest.

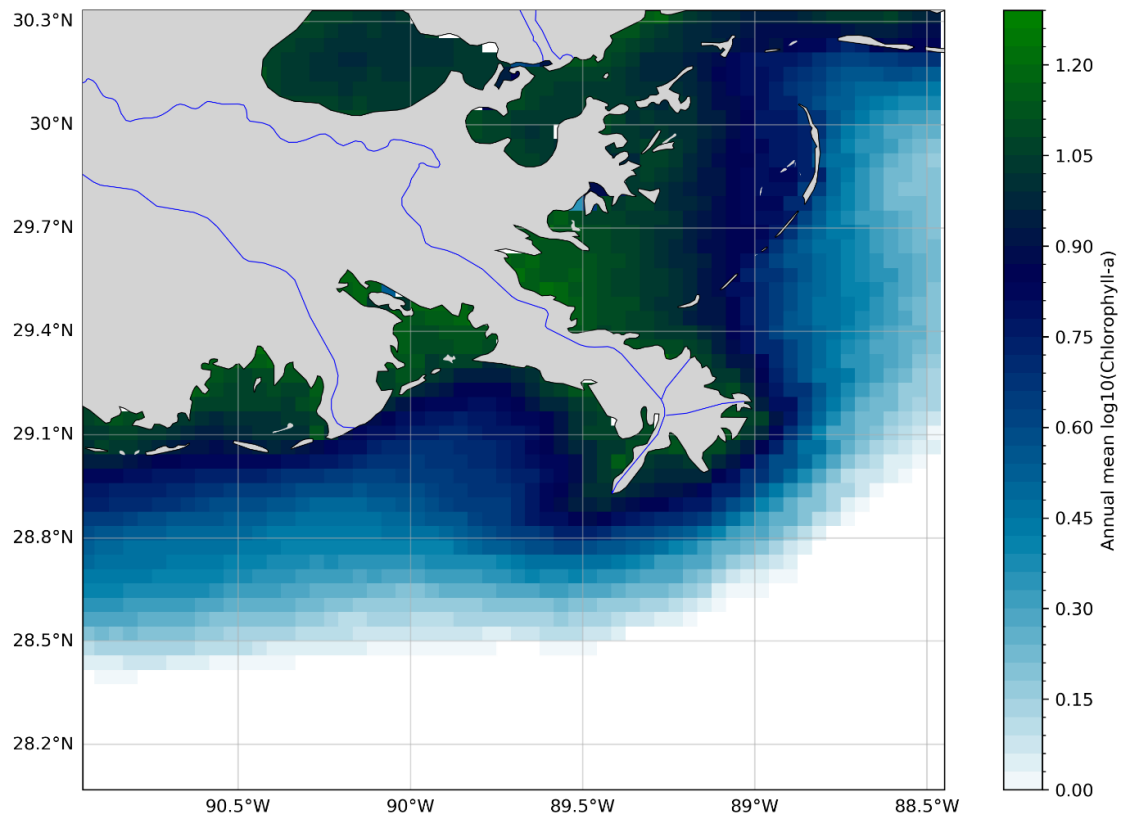


Figure 3: Logarithmic converted annual mean chlorophyll-a concentrations ($\text{mg} \cdot \text{mm}^{-3}$) along the Louisiana, US coastline.

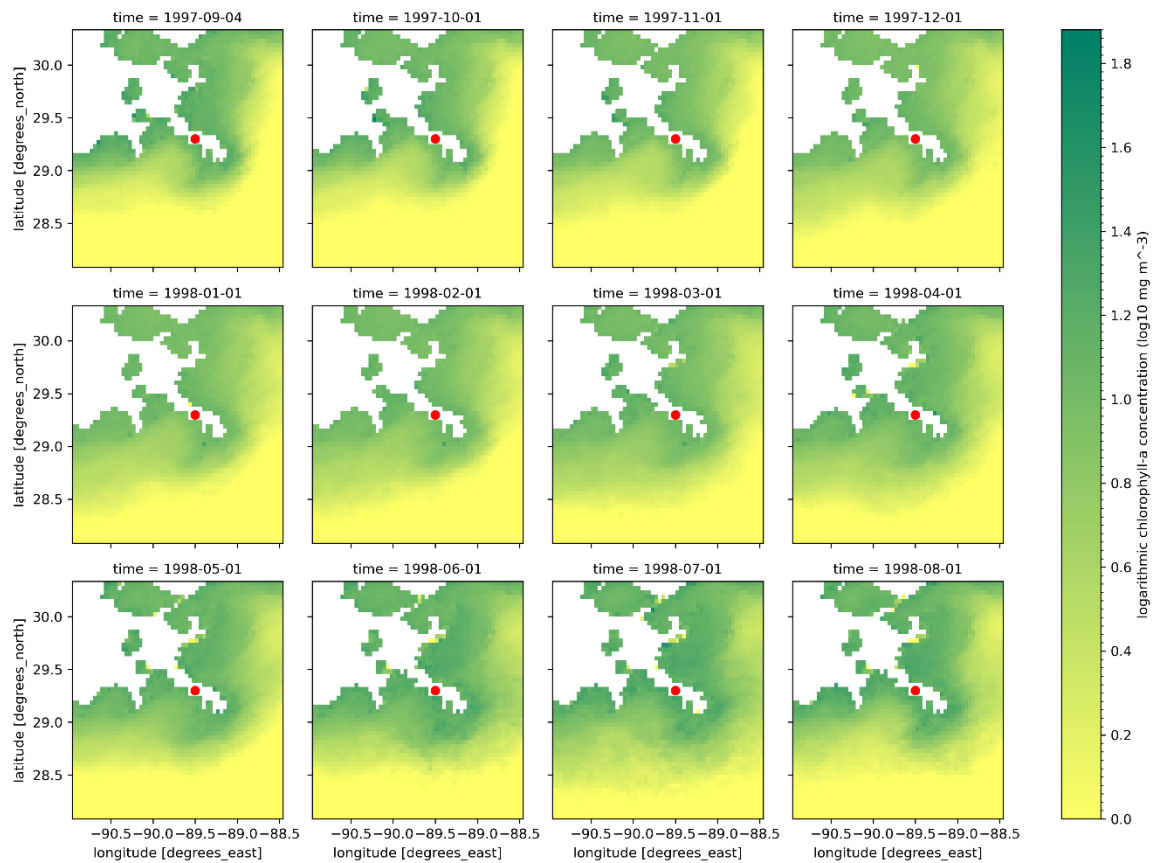


Figure 4: Monthly snapshots of the mean chlorophyll-a concentrations to show seasonal variations. Location of point of interest is indicated.

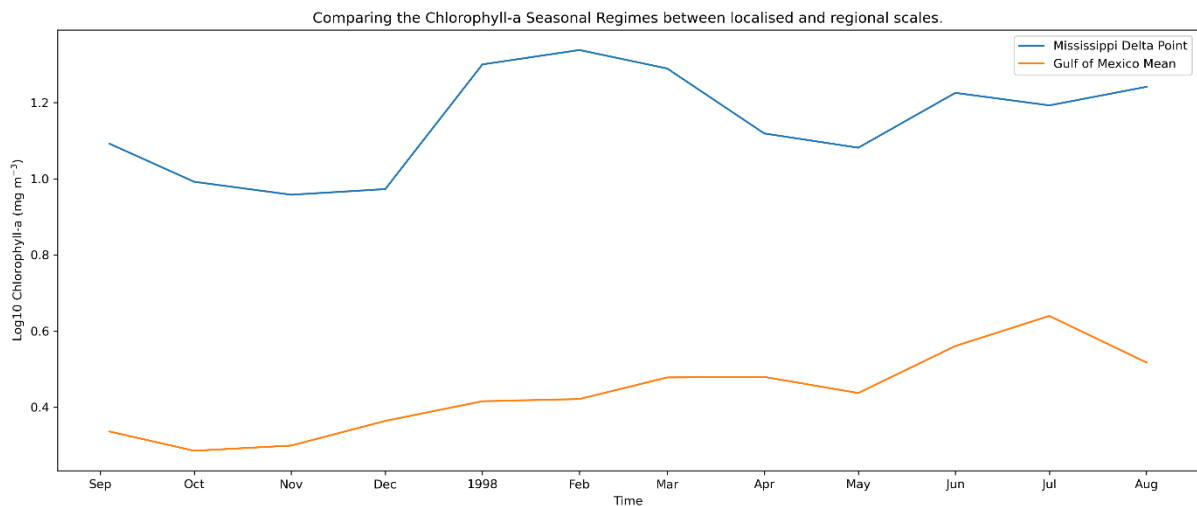


Figure 5: Comparing the climatological timeseries of the greater Gulf-of-Mexico average chlorophyll-a concentration to that of a single point of high concentration adjacent to the Mississippi Delta.

Concluding thoughts:

The data provided insights on the chlorophyll-a conditions on a monthly, seasonal and annual basis over the same period one of the strongest El Niño events occurred over. That overlap can not be shown with the current dataset to see how the event may have affected concentrations before and after the period of interest. However, we now know that variations over months and seasons were not visibly determinable and other tests would need to be conducted to then confirm whether any statistical differences exist between chlorophyll-a concentrations between months and therefore seasons.

5. References:

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6. Report on how the Large Language Model (LLM), Claude was used:

Topic:	What AI was used for:	Prompt examples:
1. Git Tracking Support	Explained how to track and commit the NetCDF file using Git despite UTF-8 warnings.	'I am trying to add it to my repository folder and tracking my changes. But I am met with an error saying the file is not encoded in UTF-8.'
2. GMT Grid file	Explain how to import a gridded file that was downloaded from GMRT.	'I have a .grd file that I downloaded and I would like to know how I can

		open this file on python to then view it'
3. Surface current context	Explain briefly what type of oceanographic processes occur in the greater Gulf-of-Mexico and locally off the coast of Louisiana, US.	'what type of surface water major current flows in the Gulf of Mexico Mississippi'
4. Grammar enhancements	Suggest improvements around grammar of introduction section	'improve the scientific grammar of the following insert but do not add any new information to it'